Computational evidence for the optimality of Conway–Kochen's 31-vector Kochen–Specker set

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We report a systematic computational search for Kochen–Specker (KS) sets with fewer than 31 vectors in dimension 3, where the Conway–Kochen construction (CK-31) has stood as the smallest known for over three decades and the best proven lower bound is 24 vectors. No sub-31 KS set is found by any of six complementary strategies: SAT-based minimization, cross-pool mixing, expanded coordinate alphabets, numerical optimization, exhaustive criticality testing, and triad density analysis. A notable empirical finding is the *norm-2 boundary*: all tested integer-like alphabets lacking the coordinate ± 2 (such as $\{0, \pm 1, \pm 3\}$) are not KS-uncolorable, consistent with the hypothesis that the cancellation identity $1 + 1 = 2$ is necessary. We establish that CK-31 is 8-critical by exhaustive enumeration of all $\binom{31}{k}$ subsets for $k = 1, \ldots, 8$ (totaling 11,460,647 SAT checks); no uncolorable subset is observed in 10^6 uniform random samples for each $k = 9, \ldots, 12$. Vertex merging produces 394 abstract KS-uncolorable graphs on 30 vertices (every non-orthogonal pair merge preserves KS-uncolorability); we resolve their realizability within the integer pool via finite constraint satisfaction (all 394 unrealizable within the $\{0, \pm 1, \pm 2\}$ pool).

INTRODUCTION

The Kochen–Specker (KS) theorem [1] demonstrates that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories. In dimension 3, the theorem requires exhibiting a finite set of rays (directions) such that no consistent $\{0, 1\}$-valued coloring assigns exactly one 1 per orthonormal basis while respecting pairwise orthogonality exclusion. The minimum number of rays needed for such a construction is a fundamental open problem.

Conway and Kochen's 31-vector set [2] uses integer coordinates from the *coordinate alphabet* $\{0, \pm 1, \pm 2\}$—the finite set of values from which ray coordinates are drawn (called "vector components" by Pavicic et al. [6]; the alphabet can be read directly from the coordinate labels on the cube diagrams used to display such constructions [5]). Two rays are orthogonal when $\sum_k u_k v_k = 0$; for coordinates from a finite alphabet, this requires an algebraic relation—a *cancellation identity*—among alphabet elements. For $\{0, \pm 1, \pm 2\}$, the key identity is $1 + 1 = 2$, yielding cancellations such as $1{\cdot}1 + 1{\cdot}1 + (-1){\cdot}2 = 0$. These cancellations create exact orthogonal triples (bases), and when enough bases interlock, no consistent $\{0, 1\}$-coloring exists. CK-31 has stood as the smallest known KS set in dimension 3 for over three decades. The best lower bound is 24 vectors, established independently by Li, Bright, and Ganesh [3] and by Kirchweger, Peitl, and Szeider [8] using SAT-based methods combined with algebraic realizability checking. (Cabello's recent 33-vector Eisenstein construction [9], the "simplest" KS set by the criterion of fewest bases, provides an independent upper bound at 33 for complex coordinates.) Closing the gap between 24 and 31—or proving that 31 is optimal—remains open.

We identify six algebraic number rings known to support KS constructions in dimension 3, each with a distinct cancellation identity producing a distinct ray pool (Table I). Rather than searching over arbitrary ray configurations, we systematically construct alphabets from these algebraic number fields and test every known algebraic source of KS orthogonalities.

METHODS

A *ray pool* is a finite set of projectively distinct rays in $\mathbb{R}^3$ (or $\mathbb{C}^3$), canonicalized so the first nonzero coordinate is positive and coordinates are gcd-reduced. A *triad* is an unordered triple of mutually orthogonal rays (an orthonormal basis up to normalization). The *orthogonality graph* of a pool has rays as vertices and edges between orthogonal pairs; triads are its 3-cliques. Throughout, merging, deletion, and embedding operations act on orthogonality graphs and their triads.

We encode KS-uncolorability as a SAT instance. For a ray set with orthogonal pairs E and triads $\mathcal{T}$, introduce Boolean variables b_v (ray v colored 1). For each triad $\{a, b, c\} \in \mathcal{T}$, add an exactly-one constraint: $(b_a \vee b_b \vee b_c)$ and $(\neg b_\alpha \vee \neg b_\beta)$ for each pair $\alpha \neq \beta \in \{a, b, c\}$. For each orthogonal pair $(u, v) \in E$ not sharing a triad, add $(\neg b_u \vee \neg b_v)$. This extends the standard KS constraint (exactly one per complete basis) by enforcing mutual exclusion on orthogonal pairs that do not complete to any basis in the pool—a strictly stronger condition that is standard in computational KS studies. UNSAT means no valid $\{0, 1\}$-coloring exists.

We tested six complementary strategies, each designed to probe a different region of the search space.

Strategy 1: SAT-verified minimization. For each of the six algebraic ray pools (Table I), we applied three heuristic methods: (a) clause-level MUS extraction,

(b) greedy deletion with 2000 random trials plus deterministic orderings, (c) swap-based local search. The SAT encoding uses Glucose4 [4] via PySAT.

Strategy 2: Cross-pool mixing. We merged rays from different algebraic pools, deduplicated by canonical form, and tested whether cross-pool orthogonalities enable smaller KS sets. All 15 pairwise combinations and the full 477-ray union of all six pools were tested with 500 greedy trials each.

Strategy 3: Larger alphabets. We tested 30 expanded alphabets: integer $\{0, \pm1, \ldots, \pm k\}$ for $k = 3, 4, 5$; extended Peres, Eisenstein, $\mathbb{Z}[\sqrt{-2}]$, and Heegner-7 pools; mixed-field alphabets; completion-expanded pools; and Pisot-number fields (plastic, tribonacci, silver ratios).

Strategy 4: Numerical optimization. Four non-algebraic methods: simulated annealing ($n = 28$–30 rays, 2×10^5 steps), CK-31 perturbation (random ray replacement, 1000 attempts), orthogonal completion from random seeds, and soft-tolerance annealing.

Strategy 5: CK-31 criticality. We tested whether CK-31 can be reduced by removing k vectors for $k = 1, \ldots, 12$, exhaustively checking all $\binom{31}{k}$ subsets for $k \leq 8$ and sampling 10^6 random subsets for $k = 9, \ldots, 12$.

Strategy 6: Triad density analysis. For each pool, we sampled random subsets of varying size and determined the threshold cardinality at which KS-uncolorable subsets first appear. This characterizes the "rarity" of KS sets within their own pools.

RESULTS

SAT-verified minimization (Strategy 1). We certify the minimum for each pool using an OCUS (Optimal Constrained Unsatisfiable Subset) procedure: for each target size $n < n_{\min}$, we add a cardinality constraint to the SAT encoding and use a CEGAR loop—when the solver finds a valid coloring, we extract a blocking clause and iterate until UNSAT. This exhaustive procedure proves that *no* KS-uncolorable subset smaller than the reported minimum exists within any of the six pools. The first five pools are certified in under 4 s each (e.g., Integer: $n = 30$ exhausted in 272 iterations, 0.1 s); the Golden pool required 18,783 iterations (303 s) at $n = 51$. As independent confirmation for the integer pool, 1,000 MUS extractions all returned subsets of size 31, yielding 162 distinct minimal KS sets.

Cross-pool mixing (Strategy 2). Despite significant cross-pool orthogonalities (e.g., Integer + Golden shares 78 cross-triads), every minimized set draws rays from a single pool only: any combination including Integer yields 31 (all Integer); without Integer, 33. The 477-ray union minimizes to 31 Integer rays. Note that greedy deletion is biased toward large basins of attraction, so mixed-pool solutions cannot be ruled out.

TABLE I. Minimum KS subset size per pool, certified by OCUS (Strategy 1). "SAT" is the OCUS-certified exact minimum within each pool. "Cross" is the minimum found when that pool is included in any pairwise or full cross-pool merge (Strategy 2).

Pool	Rays	Triads	SAT	Cross	Larger
Integer	49	26	**31**	31	31
Eisenstein	57	22	33	33	33
Peres	49	16	33	33	33
$\mathbb{Z}[\sqrt{-2}]$	49	16	33	33	33
Heegner-7	145	42	43	43	43
Golden ratio	205	166	52	52	52

TABLE II. Selected expanded alphabets (Strategy 3). "KS" indicates whether the pool is KS-uncolorable; "Min" is the best greedy result in 300–500 trials.

Alphabet	Rays	Min	Note
$\{0, \pm1, \pm2\}$	49	31	Baseline (CK-31)
$\{0, \pm1, \pm2, \pm3\}$	145	31	Absorbs to CK-31
$\{0, \pm1, \pm2, \pm3, \pm4\}$	289	36	Heuristic limit
$\{0, \pm1, \pm3\}$	49	—	Not KS
$\{0, \pm1, \pm4\}$	49	—	Not KS
Eisenstein c=3, n≤6	345	31	Absorbs to CK-31
Peres + ± 2	109	31	Absorbs to CK-31
Heegner-7 + ± 2	253	31	Absorbs to CK-31
Golden + ± 2	637	33	No absorption
$\{0, \pm1, \pm2, \pm i\}$	127	31	Mixed-field

Larger alphabets (Strategy 3). We tested 30 expanded alphabets (Table II). Every extension that includes ±2 converges to 31 using integer rays. Greedy minimization is heuristic, so for large pools (e.g., $\{0, \pm1, \ldots, \pm4\}$ with 289 rays reaching only 36 in 500 trials) the true minimum may be lower; however, every pool containing the $\{0, \pm1, \pm2\}$ subset found 31 when given sufficient trials.

The norm-2 boundary. The alphabets $\{0, \pm1, \pm3\}$ and $\{0, \pm1, \pm4\}$—which lack ±2—are *not KS-uncolorable at all*, despite generating 49 rays with 10 triads each. This is consistent with the hypothesis that the norm-2 cancellation identity $1 + 1 = 2$, which enables exact orthogonalities such as $1{\cdot}1 + 1{\cdot}1 + (-1){\cdot}2 = 0$, is necessary for KS-uncolorability from integer-like alphabets. More broadly, across all tested fields, KS-uncolorability is observed only when one of two cancellation mechanisms is present: *norm-2 cancellation* (an identity producing the value 2 from products of ring elements, as in $1 + 1 = 2$, $(\sqrt{2})^2 = 2$, $\alpha\bar{\alpha} = 2$) or *phase cancellation* (a root-of-unity sum $1 + \omega + \omega^2 = 0$). In all tested rings where neither mechanism is available—including those with generator norm ≥ 3—the pool is colorable despite containing orthogonal triples. Three higher-degree fields (plastic ratio, tribonacci, silver ratio) also produce only colorable pools, extending this observation beyond quadratic fields. The boundary is empirical: multi-term cancellations not involving 2 may exist in untested fields.

Numerical optimization (Strategy 4). All four non-algebraic methods fail: simulated annealing from scratch produces at most 5–6 orthogonal pairs and zero triads (2×10^5 steps); CK-31 perturbation (removing 1–3 rays and adding random replacements, 1000 attempts each) never restores uncolorability; orthogonal completion from random seeds achieves at most 1 triad. These failures are unsurprising—exact orthogonality is a measure-zero condition in continuous space—and primarily motivate the algebraic/discrete approach.

We also tested whether minimal KS sets can be compressed by *merging* non-orthogonal ray pairs—identifying two vertices u, v into a single vertex that inherits the union of their edges and triad memberships (triads that would contain both u and v are discarded as degenerate; duplicate triads are merged). This property is *universal*: across all six algebraic islands, every non-orthogonal pair merge preserves KS-uncolorability—394 merges (CK-31), 456 (Peres-33), 450 (Eisenstein-33), 456 ($\mathbb{Z}[\sqrt{-2}]$-33), 796 (Heegner-7-43), 1,204 (Golden-52), for a total of 3,756 merges with 100% preservation. Intuitively, merging non-orthogonal vertices adds constraints (the merged vertex inherits both neighborhoods) without removing any; since KS-uncolorability is monotone under constraint addition, preservation is expected—though not guaranteed, since triad structure can change. We conjecture that every minimal KS set is *merge-saturated*. For the CK-31 merges specifically, Z3 [7] with nonlinear real arithmetic could not resolve their $\mathbb{R}^3$ realizability ("unknown" after 60 s per instance, 90 variables each).

To circumvent this, we reformulated realizability within the integer pool as a *finite SAT encoding*. Given a candidate graph $G = (V, E)$ on n vertices, we introduce Boolean assignment variables $x_{v,r}$ (vertex v mapped to ray r, $1 \le r \le 49$) and encode three constraint families:

(C4a) *Exactly one ray per vertex:* $\bigvee_r x_{v,r}$ and $\neg x_{v,r_1} \vee \neg x_{v,r_2}$ for all $r_1 < r_2$.

(C4b) *Orthogonality:* for each edge $(i, j) \in E$ and each non-orthogonal ray pair (r_1, r_2): $\neg x_{i,r_1} \vee \neg x_{j,r_2}$.

(C4c) *Injectivity:* $\neg x_{i,r} \vee \neg x_{j,r}$ for all $i < j$, all r.

This is a *weak* (non-induced) embedding: edges must map to orthogonal ray pairs, but non-edges are unconstrained and may acquire incidental orthogonality. A satisfying assignment is a constructive embedding in $\mathbb{R}^3$; UNSAT means unrealizable within the $\{0, \pm 1, \pm 2\}$ pool (but not necessarily in $\mathbb{R}^3$ generally). For $n = 30$ with ~70 edges, this yields 1470 variables and ~218,000 clauses, resolved by Glucose4 in under 1 s per graph (Intel i5-12450H). All 394 merged graphs are *unrealizable* within the integer pool. Every possible single-merge compression of CK-31 preserves abstract KS-uncolorability, yet none can be realized with $\{0, \pm 1, \pm 2\}$ coordinates. Whether any are realizable in $\mathbb{R}^3$ with other coordinates remains open and exceeds current SMT capabilities.

CK-31 criticality (Strategy 5). CK-31 is k-critical for all tested k up to 12 (Table III). The exhaustive results ($k \le 8$) cover all 11,460,647 subsets and constitute a proof of 8-criticality. The sampled results ($k = 9$–12) each test 10^6 uniformly random subsets, none of which remain KS-uncolorable. By the rule of three, the fraction of uncolorable subsets (if any exist) is bounded by $p_k \lesssim 3 \times 10^{-6}$ at 95% confidence, assuming uniform sampling; however, uncolorable subsets may be concentrated in specific structural regions, so this bound is conservative. At $k = 7$, removing any 7 of 31 rays leaves only 24—the Li–Bright–Ganesh lower bound—yet every one of the 2,629,575 resulting 24-ray subsets is colorable. This establishes that no 24-ray sub-configuration of CK-31 is a KS set.

Triad density threshold (Strategy 6). In 500 uniform random samples per size (an exploratory diagnostic, not a precise threshold estimate), no KS-uncolorable subset of the integer pool is observed below $n = 42$ out of 49 rays (upper bound on prevalence: $p \lesssim 0.006$ at 95%); for the Eisenstein pool, $n = 48$ out of 57; for Heegner-7, none at any sampled size up to $n = 75$ out of 145. Among all tested constructions, CK-31's minimum achieves the highest pair density (15.3% of $\binom{31}{2}$ pairs are orthogonal). These results show that triad count alone does not determine uncolorability—the topological interlocking of constraints matters.

DISCUSSION

Our results establish four structural facts about KS sets in dimension 3.

First, *every pool minimum in Table I is the exact minimum within its pool*: the OCUS procedure exhaustively certifies all six entries. The first five pools are certified in

TABLE III. Extended criticality of CK-31. For $k \le 8$, all $\binom{31}{k}$ subsets tested exhaustively; for $k \ge 9$, 10^6 random subsets sampled.

k	Remain	Subsets	Method	Result
1	30	31	Exhaustive	All colorable
2	29	465	Exhaustive	All colorable
3	28	4,495	Exhaustive	All colorable
4	27	31,465	Exhaustive	All colorable
5	26	169,911	Exhaustive	All colorable
6	25	736,281	Exhaustive	All colorable
7	24	2,629,575	Exhaustive	All colorable
8	23	7,888,725	Exhaustive	All colorable
9	22	2.0×10^7	Sampled	All colorable
10	21	4.4×10^7	Sampled	All colorable
11	20	8.5×10^7	Sampled	All colorable
12	19	1.4×10^8	Sampled	All colorable

under 4 s each; the Golden pool ($n = 51$) required 18,783 CEGAR iterations (303 s). Combined with the rigidity of CK-31 [10] (uniqueness up to unitary equivalence), this certifies 31 as optimal within integer coordinates and 33 as optimal within Peres, Eisenstein, and $\mathbb{Z}[\sqrt{-2}]$.

Second, *the known algebraic pools are self-contained*: cross-pool mixing never produces a smaller KS set than what exists within a single pool.

Third, *CK-31 is deeply critical.* The 8-critical result (exhaustive) means that CK-31 cannot be reduced by deletion of *any* 8 or fewer rays—this exhaustive test covers all 11.5 million subsets. Even at $k = 7$, where 24 rays remain (the current lower bound), every subset is colorable.

Fourth, *density arguments are insufficient for a proof of optimality.* KS sets are extraordinarily rare even within pools that contain them. The uncolorability depends on the precise topological interlocking of basis constraints—a property that cannot be captured by counting triads or pairs alone. This suggests that closing the 24–31 gap will require realizability-based arguments (extending the Li–Bright–Ganesh framework) rather than combinatorial density bounds.

Working in $\mathbb{C}^3$ does not help: our complex pools all minimize to ≥ 33.

The vertex-merging experiment reveals the nature of the obstacle: all 3,756 merges across six islands preserve abstract KS-uncolorability yet none of CK-31's 394 merges are realizable within the integer pool. This universal merge saturation—observed for every minimal KS set in every tested algebraic island—suggests a structural property: no vertex is expendable enough that merging it destroys the constraint structure. Within the integer coordinate framework, the 24–31 gap is a realizability problem: combinatorially viable 30-vertex KS graphs exist in abundance, but none embed in the known algebraic pools.

We do not prove that 31 is optimal. Among all tested two-element alphabets of the form $\{0, \pm 1, \pm x\}$, the cancellation identities producing KS-uncolorable pools are $x = 2$, $|x|^2 = 2$, $|x|^2 = x + 1$, and $x^2 + x + 1 = 0$—corresponding to the six pools in Table I, none achieving fewer than 31. An exploratory *density trend* is observed: simpler cancellation identities produce denser orthogonality structures (integer: 0.53 triads/ray, minimum 31; Peres: 0.45, minimum 33; golden ratio: 0.20, minimum 52). This trend suggests—but does not prove—that achieving sub-31 density from a more complex identity would be difficult. The principal untested territory is multi-element alphabets from higher-degree number fields, where qualitatively new cancellation mechanisms could arise; our tests of Pisot-number fields (plastic, tribonacci, silver ratios) found no KS-uncolorable pools, but these do not exhaust the possibilities.

Within the algebraic framework tested, every pool minimum is OCUS-certified (Table I). The Cortez–Morales–Reyes $N(S) = \mathrm{lcm}\{\|v\|^2 : v \in S\}$ invariant [11] provides an independent algebraic perspective: CK-31 has $N = 30$ ($= 2 \times 3 \times 5$), while the Eisenstein island achieves $N = 6$—exactly the minimal ring $\mathbb{Z}[1/6]$ of their theorem. Every island requires primes 2 and 3; the additional prime 5 in CK-31 arises from its $\|v\|^2 = 5$ rays. A sub-31 KS set could in principle exist in an untested number field with a qualitatively new cancellation identity. (Note that "smallest" is not the only measure of simplicity: Cabello [9] argues that his 33-vector Eisenstein set is "simplest" by fewest bases and highest symmetry. Our focus on vertex count is complementary.)

These results are directly applicable to the realizability-based lower-bound programs of Li, Bright, and Ganesh [3] and Kirchweger et al. [8], both of which enumerate abstract KS hypergraphs on n vertices and check whether they can be realized as rays in $\mathbb{R}^3$. Both programs are bottlenecked by Z3 returning "unknown" on embeddability queries above order 20—a limitation we also encountered. Our finite pool SAT encoding (C4a–C4c) offers a complementary approach: by reformulating realizability within a finite ray pool as a pure Boolean satisfiability problem (rather than nonlinear real arithmetic), it resolves instances that defeat Z3. A SAT result provides a constructive $\mathbb{R}^3$ embedding with explicit coordinates, short-circuiting the expensive Z3 check; UNSAT means only that no embedding exists within the tested pool, and the candidate must fall through to Z3. The encoding is fast (under 1 s per 30-vertex graph) and could serve as a "quick accept" screen in the LBG pipeline.

Additionally, our algebraic data suggest three empirical generation-phase observations consistent with all six tested pools: every vertex participates in at least 2 triangles, edge density $\geq 0.15 \cdot \binom{n}{2}$, and triad count $\geq \lceil n \cdot 1.45/3 \rceil$. These are not proven necessary and should be treated as heuristic pruning aids only. The 394 merged graphs also yield candidate blocked subgraphs: extracting locally minimal unrealizable subgraphs from these order-30 graphs could extend MathCheck's blocking catalogue.

Combining these algebraic and SAT-based pruning tools with realizability checking could reduce the LBG search space by orders of magnitude.

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